

The Role of the Forest Soil Propagule Bank in Assisted Natural Recovery after Oil Sands Mining

Dean D. Mackenzie^{1,2} and M. Anne Naeth¹

Abstract

This study was undertaken to determine if use of stratified organic layers of intact litter, fragmented litter, and humus on the forest floor (LFH) improves establishment of upland native boreal plant species during oil sands reclamation in Alberta, Canada. The abundance and composition of vascular plant species in the soil propagule bank were determined for LFH and peat materials before salvage from donor sites and 18 months after application on the receiver site. Applications of 10 and 20 cm were evaluated. Various soil properties were assessed to determine impacts of donor materials. In the growth chamber, LFH donor material had significantly more plant species emerge (37) from the propagule bank than did peat donor material (19). In the field, LFH treatments had significantly higher species richness (49, 47, 24, and 25 species for LFH 10 cm, LFH 20 cm, peat 10 cm, and peat 20 cm

treatments, respectively), plant abundance, and soil nutrients than peat treatments. Application thickness of peat had little effect, but 20 cm of LFH was more beneficial for plant community establishment than 10 cm. LFH treatments had narrower C:N ratios and higher soluble potassium and available phosphorus than peat. Applying 10 versus 20 cm of donor material increased admixing of fine-textured subsoil, reducing organic carbon, nitrogen, and potassium; these effects were greater for LFH than peat treatments. Thus, addition of LFH aids in creating diverse ecosystems on reclaimed upland landscapes by providing a source of propagules for revegetating upland boreal forest communities and improving nutrient availability for plants.

Key words: boreal forest, LFH, reclamation, revegetation, seed bank, species diversity.

Introduction

Open pit mining of oil sands in mixed-wood boreal forest of northeastern Alberta, Canada, creates one of the world's largest and intense disturbances, with more than 315,000 ha susceptible to this type of disturbance (Alberta Government 2005). Current regulations require that disturbances be reclaimed to diverse, self-sustaining boreal forest communities similar to those in the surrounding region (Alberta Environmental Protection 1998; Fung & Macyk 2000). The majority of mined lands were formerly peatlands but are generally reclaimed to upland forests. The peat layer and underlying mineral substrates are commonly salvaged and used as surface soil for reclamation of upland landscapes. Revegetation focuses on transplanting trees and leaving understory species to establish through natural processes. Low viable propagule abundance in bogs, propagule density dilution from stripping peat with underlying mineral soil, wind-dispersed weeds, and poor establishment of peatland species in dry upland soils make natural recovery problematic. Seeding understory species is not feasible because seed for most boreal species, with

the exception of grasses, is not commercially available (Lanoue & Qualizza 2000).

A potential source of native plant propagules exists in a thin organic horizon of fresh, intact, identifiable litter (L), fragmented and fermenting litter (F), and humus (H) with small amounts of moss in upland forest (Paré et al. 1993). The LFH horizon contains abundant propagules for revegetation after natural disturbances, such as fire and tree throw (Whittle et al. 1997), which suggests that salvaging this horizon could provide propagules of upland species tolerant of drier conditions that are not commercially available (Leck et al. 1989; Qi & Scarratt 1998).

Using donor soils for natural recovery on mine sites is well documented for subtropical, temperate, and arid regions (Iverson & Wali 1981; Koch et al. 1996; Bakker & Berendse 1999; Holmes 2001); however, its applicability in reclaiming upland boreal forest plant communities has not been well studied. The few studies assessing application thickness of topsoil on mine sites concluded that thick or thin spreading gives similar results in seedling density and species richness (Rokich et al. 2000; Zhang et al. 2001). For example, in the northern jarrah forest of Western Australia, Grant et al. (2006) found that most species germinated from depths less than 2 cm and thus, donor soil use could be maximized if spread thin.

Considerable work has been done in the boreal forest on mined peatland restoration (Giard et al. 2002; Campbell et al. 2003; Rochefort et al. 2003; Cobbaert et al. 2004).

¹ Department of Renewable Resources, University of Alberta, Edmonton, Alberta, Canada T6G 2H1

² Address correspondence to M. A. Naeth, email anne.naeth@ualberta.ca

These studies show that successful restoration of sphagnum and other fen or bog species is dependent on introduction of propagules from donor sites, creation of suitable moisture regimes for desired species, availability of nutrients, and propagule burial depth. Although not directly applicable to oil sands reclamation where the goal was to create upland forest communities with peat donor soil, the same factors required for successful peatland restoration may be applicable to establishing upland forest. Most field research, other than peatland restoration, has been small scale with small equipment or hand tools. Few studies have attempted to apply treatments using full-size mining equipment, necessary to make the practice economical.

The overall objective of this study was to determine if use of salvaged LFH material improves establishment of native boreal plant species over the traditionally used peat–mineral mix. Vascular plant establishment, species composition, species diversity, and various soil properties were assessed in response to two application thicknesses (10 and 20 cm) of two donor materials (peat and LFH materials) in both field and growth chamber environments. The field study was designed to determine plant propagule responses to field-scale equipment, realistic industrial applications, and natural environmental conditions. The growth chamber study was designed to assess potential available propagules under controlled conditions. LFH treatments were expected to have more plants and higher species richness and species diversity than peat treatments because the reclamation was occurring on upland landscapes and the salvaged LFH material originated from upland landscapes. LFH and peat were expected to affect soil properties in similar ways, increasing nutrient and organic matter content and reducing bulk density, all beneficial for plant establishment. Handling of donor materials was expected to reduce propagule emergence in the field versus the growth chamber.

Methods

Research Site Description

The research area was 40 km north of Fort McMurray, Alberta, Canada, at the Syncrude Canada Ltd. mine site (lat 56°59'N, long 111°37'W) within the central mixed-wood subregion of the boreal natural region. The climate was temperate with short, cool summers and long, cold winters (Strong & Leggat 1992). Mean annual temperature was 0.7°C and the 1971–2000 long-term average annual precipitation was 455.5 mm, with approximately 342.2 mm of rain and 155.8 cm of snow (Environment Canada 2003).

Prior to disturbance, soils were organic in lowlands and fine-textured mineral in uplands. Predisturbance vegetation was representative of the mixed-wood boreal forest. Undisturbed organic soils in this region are dominated by Black spruce (*Picea mariana* (P.) Mill.) and Tamarack

(*Larix laricina* (Du Roi) K. Koch) and mineral soils are dominated by a coniferous–deciduous forest mix. Uplands typically consist of Trembling aspen (*Populus tremuloides* Michx.), Balsam poplar (*P. balsamifera* L.), and White spruce (*P. glauca* (Moench) Voss) (Fung & Macyk 2000).

Donor Sites Description and Sampling

LFH and peat donor sites were located adjacent to each other, bordering an oil sands extraction pit. The area had been cleared of timber and drained in 2002 for mining. Prior to timber salvaging, the LFH donor site vegetation was Trembling aspen with sparse White spruce and a diverse understory of grasses, shrubs, and forbs. The peatland donor site was located in a transition zone between the upland donor site and a bog. Soil pH varied between 4.3 and 5.5; lower pH values occurred most distant from the upland forest. Mineral soil underlying the peat was fine textured. Peatland vegetation was black spruce with willow, sedge, ericaceous shrubs, and sphagnum mosses. Organic horizons at the peat donor site were greater than 40 cm and the LFH donor soil had a thin LFH layer (7.5 cm); both sites had underlying fine-textured mineral soil.

At each donor site, three 250- to 300-m-long transects were located 75 m apart; sampling locations were at 10-m intervals ($n = 84$ samples). At each sampling location, a 0.1-m² quadrat was used to determine species present and the soil propagule bank was sampled from a 10 × 10-cm area to a depth of 10 cm from the surface with a serrated knife. Samples were cut in half (0–5 and 6–10 cm) once removed from the ground then transported to a growth chamber to quantify the potential propagule bank under controlled conditions.

Receiver Site Experimental Design and Sampling

A complete randomized design that consisted of two factors (cover source and application thickness) and two levels of each factor was used. Cover sources were salvaged LFH and peat materials and application thickness of cover material was approximately 10 and 20 cm. The experiment was located on a southeast facing complex slope of a saline sodic overburden dump on mined land. The slope complex consisted of three 7–16% forward slopes interspersed with two 2–4% bench components. Treatments ran from upper- to mid-slope positions. Each treatment was applied to three 25 × 150-m areas to accommodate operational scale equipment and material availability. The research site was constructed using common reclamation practices, except for the cover materials. Suitable subsoil (90 cm of nonsaline, nonsodic overburden) was placed on top of the saline sodic overburden to buffer electrical conductivities greater than 4 dS/m and sodium adsorption ratios from 18–37. Prior to each treatment application, snow was removed and the surface smoothed with a Caterpillar D10R crawler tractor.

Peat and LFH were salvaged in November 2003 using D10 Caterpillar bulldozers. The entire peat layer with 40–60 cm of underlying mineral soil was salvaged; the entire LFH layer with 5–20 cm of underlying mineral soil was salvaged. Hereafter, these donor mixes of LFH and peat with mineral soil material are referred to as peat and LFH materials. Materials were stored in small windrows, 2–3 m high and 4–6 m wide, until treatment application.

In February 2004, donor materials were removed from the windrows using a Hitachi E2756 excavator, which placed materials into Hitachi SH4500 haul trucks for transport. Haul trucks were loaded with approximately 160 m³ of donor material. The 20-cm application treatments received five truckloads and 10-cm application treatments received three truckloads. Materials were distributed evenly as possible over the treatment areas then spread with a D10R Caterpillar crawler tractor. Applying peat material in winter resulted in an uneven application thickness due to large frozen masses of peat (1 × 2 × 2 m in size). A Caterpillar D6LPG crawler tractor was used to pull a 15-m-wide pipe over the peat treatments once the clumps were partially thawed to redistribute the peat and obtain the desired thickness.

The soil propagule bank in LFH and peat treatments was sampled after peat treatments were redistributed to determine the effect of materials handling on plant propagules. Once treatments were in place, samples were taken to quantify propagule emergence in the growth chamber every 10 m along two 140 m randomly located transects using a 7.5-cm-diameter core at a 7.5-cm depth ($n = 28$).

In each treatment, vegetation composition was assessed in late-July 2004 and 2005 using four randomly located transects. Along each transect, 0.1-m² quadrats were placed at 5-m intervals. Plant density and canopy cover by species were measured in each quadrat. Rhizomatous plants that formed clumps were considered individual plants and individual crowns along a stolon were counted as individual plants. Density of Narrow leaved hawk-beard (*Crepis tectorum* L.) was included in cover estimates when densities were greater than 100 plants per quadrat. Species nomenclature followed Moss (1983).

Soils were sampled in early August along the same transects used for vegetation assessment. Seven 0.1-m² quadrats per replicate were located along two randomly selected transects and soil samples were taken in the center of each quadrat. The homogeneous donor material layer was sampled with a shovel and the underlying 0–20 cm of subsoil was sampled with a Dutch auger. Five bulk density measurements were taken per experimental unit with a 7.5-cm-diameter core to a 7.5-cm depth.

All soil samples were analyzed according to Carter (1993). Percent saturation, pH, electrical conductivity, sodium adsorption ratio, soluble cations (calcium, potassium, magnesium, and sodium), and soluble anions (chloride and sulfate) were determined from a saturated paste extract. Total nitrogen (TN) was analyzed by digestion with a pre-treatment of Devarda's alloy to convert nitrate

to ammonium; total organic carbon (TOC) was determined by wet oxidation-redox titration. A conversion factor of 1.72 multiplied by percent TOC was used to calculate percent organic matter. Available phosphorus was analyzed using the modified Kelowna extract method. Nitrate (NO₃⁻) and ammonium (NH₄⁺) were determined using extraction with 2.0M KCL. Soil samples were oven dried at 105°C to constant mass (approximately 48 hours) and weighed to determine bulk density.

Growth Chamber Methods

Propagule emergence from donor soil treatments was quantified in a growth chamber under controlled light and temperature conditions. The 10 × 10 × 10-cm samples from the donor sites were subsampled with a 5-cm-diameter corer to accommodate growth chamber space. Samples from donor and receiver sites were thoroughly mixed by hand then spread to a thickness of 1–2 cm in plastic containers lined with 1 cm of vermiculite and placed randomly throughout the growth chamber. Soils were watered with distilled water as needed to prevent drying. Growth chamber temperatures mimicked growing conditions at the field site (21°C during the day for 16 hours and 15°C at night for 8 hours). Emerged plants were identified and counted 2 weeks after potting and at monthly intervals thereafter for 6 months. Samples were remixed after each enumeration to promote emergence by bringing up buried seeds and reducing thickness of the moss layer to promote light penetration (Thompson et al. 1997). Some species were not identifiable because of death between enumeration periods or prior to identifiable structures emerging.

Statistical Analyses

Statistical analyses were the same for growth chamber and field-collected data. Species were categorized into plant groups based on morphology (i.e., grasses, sedges, rushes, woody plants, forbs, pteridophytes), life history (i.e., perennial, annual/biennial), and origin (i.e., native to North America, introduced). Unknown monocots and dicots were only included in the total. Diversity was calculated according to the Shannon–Wiener index (H') and evenness was calculated using the formula: $E = H' / \log_{10}R$ (Magurran 1988). Sorenson's similarity index was calculated using the formula: $S = 2N / (2N + R_1 + R_2)$, where N = number of species in both sites and R_1 and R_2 were species richness in donor and receiver sites, respectively. The small number (two to five species for each experiment) of unidentified plants were excluded from diversity measures. The Mann–Whitney test was used to compare differences in abundance of plant groupings between peat and LFH donor sites for the 0- to 10-cm depth, $\alpha = 0.05$ (Zar 1999). Two-way analysis of variance was used to determine significant differences among treatments for response variables at the receiver site

(Zar 1999). Subsample data were averaged to give one value per experimental unit for each variable. Significant interaction effects were further analyzed using Tukey's post hoc test for significant differences between treatments (Zar 1999). Spearman's correlation coefficients were calculated to explore relationships among plant group canopy cover and environmental variables.

Results

Donor Site Propagule Bank (Growth Chamber)

Propagule emergence in the growth chamber from both LFH and peat materials comprised 20 families, 32 genera, and 37 species. Richness in LFH and peat treatments was 37 and 19 species, respectively. Emergence was higher for most plant groups in LFH donor material (total $U_{[85]} = 492.5$, $p = 0.0016$; grass $U_{[85]} = 599.5$, $p = 0.0134$; sedge $U_{[85]} = 498.5$, $p = 0.0012$; rush $U_{[85]} = 648.5$, $p = 0.0432$; forb $U_{[85]} = 334.5$, $p = 0.0001$; native $U_{[85]} = 508$, $p = 0.0035$; perennial $U_{[85]} = 548$, $p = 0.0106$; and annual/biennial $U_{[85]} = 511$, $p = 0.0024$) than in peat donor material except for pteridophytes, woody, and introduced species (Table 1). In both peat and LFH, 80% of plants were perennials and most plant groups had higher densities in the upper 5 cm than at 6–10 cm. Dicotyledons were most abundant in peat (45%) and monocotyledons (50%) in LFH treatments. Almost 90% of emergents were from seed in LFH compared with 60% in peat treatments.

Receiver Site Propagule Bank (Growth Chamber)

A total of 20 families, 27 genera, and 33 species (3 shrubs, 22 herbs, 3 grasses, 1 lily, 1 typha, 2 rush and 1 horsetail) emerged in the growth chamber from receiver site treatments (Table 2). LFH treatments were more similar ($F_{[1,8]} = 5.80$, $p = 0.0426$) to donor site vegetation than peat

treatments. LFH treatments had 29 species emerge and peat treatments had 16. Tickle grass (*Carex* sp., *Agrostis scabra* Willd.), Baltic rush (*Juncus balticus* Willd.), and Rough cinquefoil (*Potentilla norvegica* L.) were the most abundant species in the propagule banks. Emerged propagule density from receiver site materials was lower than that from donor materials and application thickness had more effect than material source (95% losses in LFH, 91% in peat 10 cm, and 77% in peat 20 cm) (Table 2). The majority of emergents originated from seed as greater than 99% of plant vegetative parts were lost after donor materials application.

Receiver Site Vegetation Establishment (Field)

Total plant density increased between the 6- and 18-month assessments by 22, 21, 29, and 39 times for LFH 20 cm, LFH 10 cm, peat 20 cm, and peat 10 cm treatments, respectively (data not published). The large increase was from *Crepis tectorum* that accounted for 85% of plant density for LFH treatments and 62% for peat treatments. Native plant density was highest in LFH ($F_{[1,8]} = 7.97$, $p = 0.0223$) and 20 cm treatments ($F_{[1,8]} = 12.22$, $p = 0.0081$). Sedge density was highest in LFH ($F_{[1,8]} = 20.05$, $p = 0.0021$) and 10 cm treatments ($F_{[1,8]} = 5.12$, $p = 0.0535$). Grass density was higher in 20 cm ($F_{[1,8]} = 4.26$, $p = 0.0730$) than in 10 cm treatments. Total plant cover increased 11, 19, 42, and 25 times for LFH 20 cm, LFH 10 cm, peat 20 cm, and peat 10 cm, respectively, during that period. The peat treatment had a higher pteridophyte cover ($F_{[1,8]} = 24.70$, $p = 0.0011$) than the LFH treatment.

In the field, most plant groups had higher density and cover in LFH than peat treatments (Tables 3 & 4). Significant interaction effects occurred for most plant group densities (total, $F_{[1,8]} = 4.62$, $p = 0.0638$; forb, $F_{[1,8]} = 4.70$, $p = 0.0621$; woody, $F_{[1,8]} = 16.81$, $p = 0.0034$; introduced, $F_{[1,8]} = 4.11$, $p = 0.0773$; perennial, $F_{[1,8]} = 10.12$, $p = 0.0130$; and annual/biennial, $F_{[1,8]} = 3.71$, $p = 0.0903$) and

Table 1. Mean ($\pm$ SE) plant group abundance in the growth chamber (emergents, /m²) from the upper (0–5 cm), lower (6–10 cm), and entire (0–10 cm) sampled strata of LFH and peat donor sites.

Plant Group	Peat						LFH					
	Upper	SE	Lower	SE	Entire	SE	Upper	SE	Lower	SE	Entire	SE
Total	2,511 ^a	519	1,103 ^b	334	3,614 ^B	615	5,481 ^a	953	3,627 ^b	610	9,108 ^A	1,204
Grass	424	234	238	221	662 ^B	329	1,136	505	773	334	1,909 ^A	626
Sedge	272	119	204	87	475 ^B	146	836	161	645	127	1,482 ^A	225
Rush	204	111	102	86	305 ^B	135	973 ^a	263	218 ^b	62	1,191 ^A	283
Forb	543 ^a	129	221 ^b	63	764 ^B	152	2,027	333	1,573	290	3,599 ^A	448
Woody	814	422	68	32	882	418	373	145	255	133	627	270
Pteridophyte	255	142	272	121	526	198	127	83	164	73	291	148
Native	2,274 ^a	489	1,035 ^b	333	3,309 ^B	584	5,072 ^a	888	3,281 ^b	610	8,353 ^A	1,154
Introduced	17	17	0	0	17	17	64	32	36	22	100	44
Perennial	2,240 ^a	484	967 ^b	334	3,207 ^B	578	4,590 ^a	850	2,754 ^b	493	7,344 ^A	1,077
Annual/biennial	51	28	68	40	119 ^B	53	545	128	564	259	1,109 ^A	304

Lowercase superscript letters within a row denote significant differences between upper and lower strata within peat or LFH treatments at $\alpha = 0.10$. Uppercase superscript letters within a row denote significant differences in the entire strata between peat and LFH treatments at $\alpha = 0.10$.

Table 2. Mean ($\pm$ SE) number of plants (emergents, /m²) in each plant group in the growth chamber from peat and LFH propagule banks at the receiver site.

Plant Group	Peat				LFH			
	10 cm		20 cm		10 cm		20 cm	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Total	309 ^b	75	829 ^a	164	342 ^b	136	527 ^a	234
Grass	51	19	178	133	78	39	89	12
Sedge	210 ^{ab}	73	552 ^a	89	127 ^b	59	226 ^{ab}	110
Rush	3	3	19	12	48	45	43	24
Forb	43	5	65	23	86	32	151	78
Woody	3	3	—	—	—	—	11	11
Pteridophyte	—	—	8	5	3	3	3	3
Native	291 ^b	71	807 ^a	166	318 ^b	112	506 ^a	237
Introduced	11	3	11	3	5	5	3	3
Perennial	288 ^b	68	788 ^a	154	288 ^b	127	444 ^a	203
Annual/biennial	14	5	30	14	35	24	65	33

Different lowercase superscript letters within a row denote significant treatment differences at $\alpha = 0.10$.

cover (total, $F_{[1,8]} = 174.36$, $p = 0.0001$; forb, $F_{[1,8]} = 256.93$, $p = 0.0001$; grass, $F_{[1,8]} = 4.33$, $p = 0.0709$; woody, $F_{[1,8]} = 33.22$, $p = 0.0001$; native, $F_{[1,8]} = 11.93$, $p = 0.0086$; introduced, $F_{[1,8]} = 20.23$, $p = 0.0020$; perennial, $F_{[1,8]} = 81.33$, $p = 0.0001$; and annual/biennial, $F_{[1,8]} = 8.69$, $p = 0.0185$). Post hoc Tukey's analysis revealed the LFH 20 cm treatment had highest densities of total, forb, grass, woody, introduced, perennial, and annual/biennial plant groups. The LFH 10 cm treatment had greater woody densities ($p < 0.1$) than the peat 20 cm but not the peat 10 cm treatment. The LFH 20 cm and LFH 10 cm treatments had greater canopy cover ($p < 0.1$) than peat treatments for total, forb, grass, native, perennial, and annual/biennial plant groups. The LFH 10 cm treatment had greater woody plant cover ($p < 0.1$) than the peat 20 cm treatments but not the peat 10 cm.

Eighteen months after treatment establishment, 64 vascular plant species comprised 39 herbs, 9 shrubs, 9 grasses, 2 trees, 2 horsetails, 1 lily, 1 typha, 1 rush, and a few unidentified willow and sedge species. More spe-

cies established and most diversity indices were higher in the LFH treatment (Table 5). Interaction effects occurred for transect richness ($F_{[1,8]} = 41.22$, $p = 0.0002$). Post hoc Tukey's analysis revealed the LFH 20 cm treatment had the greatest richness and the LFH 10 cm treatment had higher values than peat treatments. The source of donor material had the most impact on diversity measures. LFH treatments had higher values for treatment richness ($F_{[1,8]} = 88.67$, $p = 0.0001$) and the Shannon–Wiener index ($F_{[1,8]} = 13.24$, $p = 0.0066$). LFH treatments had a greater Sorenson similarity index when comparing species established at the research site to species in the respective donor site propagule bank ($F_{[1,8]} = 11.32$, $p = 0.0099$) and to the respective donor site aboveground vegetation ($F_{[1,8]} = 41.21$, $p = 0.0002$). Peat treatments had greater evenness ($F_{[1,8]} = 5.23$, $p = 0.0515$) than LFH treatments. A high proportion of species tolerant to a wide range of growing conditions and those that disperse seed through wind were found in peat treatments.

Table 3. Mean ($\pm$ SE) percent cover of plant groups in LFH and peat treatments 18 months after application of donor material.

Plant Group	LFH				Peat			
	20 cm	SE	10 cm	SE	20 cm	SE	10 cm	SE
Total	35.7 ^a	0.7	20.1 ^b	1.0	5.0 ^c	0.3	6.1 ^c	0.3
Grass	2.6 ^a	0.1	1.6 ^b	0.3	0.6 ^c	0.2	0.4 ^c	0.1
Sedge	1.1	0.3	0.6	0.2	0.6	0.0	0.4	0.2
Rush	—	—	—	—	0.03	0.0	—	—
Forb	28.9 ^a	0.3	16.6 ^b	1.4	3.3 ^c	0.1	4.3 ^c	0.4
Woody	3.0 ^a	0.4	1.0 ^b	0.1	0.1 ^c	0.1	0.5 ^{bc}	0.2
Pteridophyte	0.1 ^b	0.1	0.2 ^b	0.1	0.6 ^a	0.1	0.6 ^a	0.1
Native	23.7 ^a	2.0	15.3 ^b	1.0	3.7 ^c	0.5	3.7 ^c	0.8
Introduced	11.9 ^a	1.5	4.7 ^b	0.7	1.3 ^b	0.5	2.4 ^b	0.7
Perennial	30.9 ^a	1.0	16.6 ^b	1.3	3.1 ^c	0.3	4.0 ^c	0.5
Annual/biennial	4.7 ^a	0.3	3.4 ^b	0.3	1.9 ^c	0.1	2.0 ^c	0.2

Different lowercase superscript letters within rows denote significant differences in canopy cover or density among treatments at $\alpha = 0.10$.

Table 4. Mean ($\pm$ SE) density of plant groups (plants, /m²) in LFH and peat treatments 18 months after application of donor material.

Plant Group	LFH				Peat			
	20 cm	SE	10 cm	SE	20 cm	SE	10 cm	SE
Total	369 ^a	42	182 ^b	65	74 ^b	19	59 ^b	6
Grass	4 ^a	1	2 ^b	0	2 ^a	1	1 ^b	0
Sedge	3 ^c	0	2 ^d	0	12 ^a	1	7 ^b	3
Rush	—	—	—	—	1	0	—	—
Forb	359 ^a	43	177 ^b	65	54 ^b	18	45 ^b	8
Woody	2 ^a	0	1 ^b	0	0 ^c	0	1 ^{bc}	0
Pteridophyte	1 ^b	0	1 ^b	1	5 ^a	1	6 ^a	1
Native	39 ^a	2	24 ^b	1	26 ^b	2	20 ^b	5
Introduced	329 ^a	43	158 ^b	64	47 ^b	20	39 ^b	9
Perennial	44 ^a	3	24 ^b	2	24 ^b	0	20 ^b	4
Annual/biennial	324 ^a	44	158 ^b	64	50 ^b	19	39 ^b	8

Different lowercase superscript letters within rows denote significant differences in canopy cover or density among treatments at $\alpha = 0.10$.

Receiver Site Soils

Soil chemical and physical properties varied among treatments. Some soil chemical properties had interaction effects including TN ($F_{[1,8]} = 3.95$, $p = 0.0822$), organic carbon ($F_{[1,8]} = 5.02$, $p = 0.0553$), organic matter ($F_{[1,8]} = 5.17$, $p = 0.0526$), and electrical conductivity ($F_{[1,8]} = 6.18$, $p = 0.0377$). TN was lowest in the LFH 10 cm treatment ($p < 0.1$). Peat treatments had more organic carbon ($p < 0.1$) and organic matter than LFH treatments, and the LFH 20 cm treatment had more carbon ($p < 0.1$) than the LFH 10 cm treatment (Table 6). The LFH 10 cm treatment had a higher electrical conductivity than the peat treatments and the LFH 20 cm treatment.

LFH treatments had higher concentrations of available phosphorus ($F_{[1,8]} = 17.96$, $p = 0.0010$) and soluble potassium ($F_{[1,8]} = 43.46$, $p = 0.0002$) and higher pH ($F_{[1,8]} = 4.53$, $p = 0.0660$) and more surface mineral soil ($F_{[1,8]} = 34.94$, $p = 0.0004$). LFH treatments had a lower carbon to nitrogen ratio ($F_{[1,8]} = 17.80$, $p = 0.0029$) than peat treatments. The 10 cm treatments had less available potassium ($F_{[1,8]} = 4.93$, $p = 0.0572$) and higher pH ($F_{[1,8]} = 11.04$, $p = 0.0105$) than 20 cm treatments.

In LFH treatments, plant growth was more responsive to all environmental variables. Soil physical properties

had stronger correlations with plant group cover than soil chemical properties in LFH treatments. An increase in percent mineral soil resulted in reduced plant densities for most plant groups in both donor materials and correlation between total plant density and percent mineral soil was stronger in LFH treatments ($r = -0.48$, $p < 0.01$) than peat treatments ($r = -0.17$, $p < 0.01$). Percent mineral soil cover was negatively correlated with vegetation cover and LFH had twice as strong a relationship ($r = 0.57$, $p < 0.01$) as peat. Percent organic material on the soil surface had the greatest correlation with total vegetation cover in LFH treatments ($r = 0.61$, $p < 0.01$).

Discussion

Propagule Banks

One comparable study in Canada had similar propagule abundance to the LFH donor site, others were much lower (Archibold 1979; Fyles 1989; Qi & Scarratt 1998). In other Canadian studies, peat propagule abundance was higher than in this study (Moore & Wein 1977). Past disturbance history (e.g., clear cutting and draining), variations in annual seed production (Qi & Scarratt 1998), and

Table 5. Mean ($\pm$ SE) species diversity in peat and LFH at the receiver sites 18 months after application of donor material.

	LFH				Peat			
	20 cm	SE	10 cm	SE	20 cm	SE	10 cm	SE
Transect richness	38.00 ^a	1.15	34.33 ^a	1.45	17.33 ^b	2.67	17.33 ^b	2.33
Treatment richness	49.00 ^a	1.53	47.00 ^a	2.65	24.33 ^b	1.86	24.67 ^b	1.98
Diversity (cover)	2.31 ^a	0.09	2.49 ^a	0.05	2.14 ^b	0.13	2.04 ^b	0.04
Evenness (cover)	0.64 ^b	0.02	0.71 ^b	0.02	0.76 ^a	0.04	0.72 ^a	0.04
Similarity P	0.34 ^a	0.01	0.35 ^a	0.01	0.31 ^b	0.02	0.31 ^b	0.01
Similarity V	0.32 ^a	0.00	0.32 ^a	0.01	0.22 ^b	0.01	0.23 ^b	0.03
Similarity V (peat versus LFH)	—	—	—	—	0.24	0.06	0.22	0.04
Similarity P (peat versus LFH)	—	—	—	—	0.30	0.03	0.29	0.01

Different lowercase superscript letters within rows denote significant treatment differences at $\alpha = 0.10$. V, aboveground vegetation; P, soil propagule bank.

Table 6. Mean ($\pm$ SE) soil parameters for the surface LFH and peat treatments 18 months after application of donor material.

	LFH				Peat			
	20 cm	SE	10 cm	SE	20 cm	SE	10 cm	SE
Potassium (mg/kg)	9.31 ^a	0.31	8.00 ^b	0.51	5.10 ^c	1.17	3.45 ^d	0.19
Phosphorous (mg/kg)	4.38 ^a	0.58	4.75 ^a	0.52	2.4 ^b	0.30	2.25 ^b	0.31
pH	6.4 ^c	0.1	6.8 ^a	0.1	6.2 ^d	0.1	6.6 ^b	0.2
EC (dS/m)	0.9 ^b	0.1	1.3 ^a	0.1	0.8 ^b	0.1	0.8 ^b	0.1
SAR	0.66	0.14	0.94	0.18	1.01	0.06	0.85	0.08
Nitrate (mg/kg)	3.79	0.29	3.34	0.30	4.20	0.64	3.89	1.18
Ammonium (mg/kg)	5.50 ^a	0.25	4.26 ^b	0.45	5.47 ^a	0.25	4.80 ^b	0.44
TN (%)	0.29 ^a	0.02	0.16 ^b	0.02	0.33 ^a	0.03	0.33 ^a	0.04
Organic matter (%)	9.7 ^b	0.6	5.7 ^c	0.6	15.9 ^a	1.3	15.4 ^a	0.1
TOC (%)	5.6 ^b	0.4	3.3 ^c	0.3	9.2 ^a	0.7	8.9 ^a	0.1
Carbon to nitrogen ratio	19.6 ^a	0.9	24.4 ^a	2.2	29.7 ^b	0.6	28.9 ^b	2.5
Mineral cover (%)	22.0 ^d	1.3	37.1 ^b	0.9	27.8 ^c	5.0	50.4 ^a	3.7
Organic cover (%)	72.2 ^a	1.4	56.8 ^c	1.2	66.8 ^b	5.2	39.5 ^d	3.4

Different lowercase superscript letters within rows denote significant treatment differences at $\alpha = 0.10$. EC, electrical conductivity; SAR, sodium adsorption ratio.

extended time in the growth chamber may explain the high-emerging propagule abundance found in this study. Vegetation and soil type play an important role in propagule bank abundance, and this study supports results from Moore and Wein (1977), where higher propagule density was found in upland deciduous forest soils than in peat organic soils.

A lower total emerged propagule abundance in the upper 7.5 cm at the receiver site compared with abundance prior to salvaging donor sites is common when soils are transferred from their original locations. Such intense disturbances favor establishment from seeds over vegetative parts (Granström 1986; Rydgren et al. 2004). Although sampling methods may not have captured all vegetative propagules, there is a very large loss. Koch et al. (1996) also found large seed density losses associated with stripping (26%), stockpiling (69%), and respreading (87%). Tacey and Glossop (1980) and Rokich et al. (2000) found deeper salvaging of donor soils significantly reduced plant density. Deep salvaging dilutes the abundant source of propagules near surface with less abundant sources below. Although the effect of stockpile overheating in temperate boreal forest has not been well researched, stockpiles in this study were moist when stripped, potentially promoting decomposition, creating anaerobic conditions, and in situ germination for light independent species that can lead to propagule death (Rokich et al. 2000). Spreading could have reduced propagule abundance, with bulldozers ripping plant propagules and reducing viability. During spreading, equipment may have redistributed the abundant source of propagules below the sampled 7.5 cm of surface soil.

The greater propagule abundance in the LFH donor site should have led to higher propagule densities at the receiver site; however, this was not the case. Furthermore, application thickness had more effect on propagule abundance than propagule source. Samples from LFH treatments and 10 cm applications had greater amounts of

mineral soil from fine particles on the surface. The mineral soil created a smooth, light impermeable layer, which may have reduced emergence. Seeds buried in high clay soil compared with depths of 1 cm may have reduced emergence due to low light penetration (Galinato & Van Der Valk 1986; Fenner 2000). Ter Heerdt et al. (1996) studied concentrated (removal of coarse and fine mineral material) and nonconcentrated clay soils and found concentrated samples had significantly higher seedling emergence. In this study, reduced emergence in LFH and 10 cm treatments resulted in lower emergence overall.

Moore and Wein (1977) found more species emerging from upland forest communities than peatlands similar to results in this study at LFH sites and treatments. The larger loss in peat treatments reflects species lost during application and a changed environment at the receiver site. Species translocated from cold, wet environments to warmer landscapes with better drainage may never establish. The cold, wet, peat environment may explain the higher contribution from plant vegetative parts compared with LFH donor soil, as unproductive soil environments can favor vegetative regenerating species (Billings & Mooney 1968; Grime 2001). A propagule pool more dependent on plant vegetative parts may be more susceptible to death compared with seed that can remain dormant for years until environmental conditions favor germination.

Field Establishment

Greater plant densities and cover in LFH than peat treatments were expected because the LFH donor site had more species adapted to mesic soil conditions and a denser soil propagule bank. Most species that established in LFH treatments, such as Strawberry (*Fragaria virginiana* Duchesne), American vetch (*Vicia americana* Muhl.), Prickly rose (*Rosa acicularis* Lindl.), and Wild red raspberry (*Rubus idaeus* L.), were better suited to mesic soil conditions than species like Labrador tea (*Ledum groenlandicum*

Oeder) and Small bog cranberry (*Oxycoccus microcarpus* Turcz.) found on organic soils. The absence of hydrophilic species, such as Baltic rush and Bog cranberry (*Vaccinium vitis-idea* L.), in both donor soils indicated drier soil conditions might prevent their emergence, considering that they were very abundant in propagule banks.

Higher plant density in LFH treatments may have been influenced by microtopographic variability relative to the smoother surface of peat treatments. LFH treatments were not smoothed after application and numerous microsites were created from woody debris and mineral clods varying in size from less than 10 cm to greater than 1 m. Although average application thicknesses were consistent throughout the general treatment area, the variable surface microtopography could have had a beneficial impact on early plant establishment. Johnson and Fryer (1992) studied effects of surface roughness and concluded seeds remained in place longer on a rough surface than a smooth surface, which enabled them to imbibe moisture for a longer time and provided a site for nutrient concentration. The significance of microsite availability was exemplified by *Crepis tectorum*. Seeds were dispersing onto the receiver site from a reclaimed area. Many seedlings established near plant bases and in dips, cracks, small rills, and near woody debris. LFH treatment microsites likely held seeds long enough to germinate and had positive effects on native species establishment. Cover reflects the amount of ground protection plants are contributing to erosion control, giving a good estimate of ecological significance and reflecting ecosystem function (Floyd & Anderson 1987). The higher cover in LFH than peat treatments can be attributed to species suited for drier conditions and greater available soil nutrients.

In other mine restoration studies, thick and thin applications resulted in similar emergence because most seeds only emerge from depths less than 2 cm (Rokich et al. 2000; Zhang et al. 2001; Grant et al. 2006). Holmes (2001) found shallow applications of topsoil resulted in higher densities of native plant species than thick applications. Soil texture was clay loam in the Holmes study, whereas in sandy soils, establishing plants may be more tolerant of admixing. Other studies established plots with smaller equipment, resulting in more uniform salvage and placement. Higher densities in the LFH 20 cm treatment resulted from less mixing with underlying subsoil during application and fewer bare areas. The higher mineral soil content at the surface of LFH 10 cm treatments may have reduced light penetration for seed germination and reduced nutrient availability for establishing plants. Water infiltration, water storage, soil nutrients, and soil stabilization also affect emergence (Bowen et al. 2005). Some studies have shown thick applications result in greater plant biomass and cover than shallow applications due to increased nutrients, organic matter, and available moisture (Pinchak et al. 1985; Bowen et al. 2005). Although results from the LFH treatments support these studies, differences between donor materials with application

thickness suggest that LFH is more responsive to application procedures than peat. If admixing can be eliminated, thin and thick applications may provide similar plant propagule abundances.

Similarities in the majority of plant group densities between the two peat application thicknesses support previous studies (Rokich et al. 2000; Zhang et al. 2001). Lack of differentiation in plant density and cover between the two peat application thicknesses indicates thin applications may provide plants with sufficient nutrients and soil moisture relative to the 20-cm industry standard application.

Species perform diverse ecological functions. Increased richness and diversity often lead to increased ecological stability, resulting in a more resilient and higher functioning plant community (Tilman 1996; Peterson et al. 1998). More species added from LFH may facilitate plant community persistence through periods of environmental stress. A diverse and species-rich plant community compared with one with a few plants will likely provide a habitat capable of servicing more pollinators and resisting more natural threats that can help maintain ecosystem functions during environmental stresses such as drought, disease, and pests. LaSalle and Gould (1993) indicate that Hymenoptera diversity is better maintained if the ecosystems are rich and diverse in vegetation. Tilman (1996) conducted a long-term study on effects of biodiversity on ecosystem stability and concluded that in drought years, plant communities with higher species numbers were more drought resistant. Although both 20 and 10 cm LFH treatments resulted in higher species richness, they had greater densities and cover of non-native species. The majority of non-native species were annuals or short-lived perennials; effects of their abundance on plant community development require further monitoring.

The community similarity between the donor materials can be explained by the low similarity of propagule banks to aboveground vegetation at the donor sites, which is common in forests (Olmsted & Curtis 1947; Kellman 1970; Thompson & Grime 1979; Qi & Scarratt 1998). Higher similarity of LFH treatments to their donor site compared with peat treatments was due to presence of species tolerant to drier soil conditions. Both donor sites were adjacent each other, and dissimilarity was anticipated to have been much greater if donor sites were not in such close proximity due to less species overlap between donor sites.

Field Site Soils

Carbon to nitrogen ratio (C:N) is commonly used as an index of soil and litter quality. The lower C:N ratio in LFH treatments may indicate a greater potential for nitrogen mineralization. Devito et al. (1999) found that peatlands had a higher C:N ratio than organic layers of a deciduous forest due to higher organic carbon. Because nitrogen is one of the major limiting plant nutrients in the boreal forest, using LFH on reclaimed landscapes may reduce dependency on nitrogen fertilizer inputs relative to

that used with a peat–mineral mix. Organic nitrogen comprises the majority of TN in this system and the lower organic matter in the LFH 10 cm treatment will limit the total pool of nitrogen that could become mineralized and available for plant uptake. Phosphorus and potassium are often limiting in upland boreal forests (Van Cleve et al. 1983). Results from this study demonstrate that the applied LFH contained greater concentrations of these nutrients relative to the peat material.

Conclusions and Reclamation Applications

The presence of propagules that will grow in a changed environment on a receptor site is important when selecting an appropriate donor soil for reclamation. Historically, the LFH horizon has not been salvaged in oil sands mining and is not used in many other disturbances. Peat or peat–mineral mixes is often the organic amendment of choice. The benefits of salvaging LFH and using it as a donor soil may outweigh the costs, especially where species are poor dispersers and are not available at a commercial scale. Adding LFH as part of a reclamation practice will result in plant communities more similar to undisturbed upland forests in a shorter time period than adding peat because plant propagules in LFH are more numerous and less hydrophilic than those in peat. Furthermore, applying thin applications of donor soils that are heterogeneous in structure to rough surfaces may reduce species composition and abundance of desired propagules for natural recovery. When applying donor soils, admixing with subsoil materials should be avoided, as it dilutes soil propagules, organic matter, and nutrients. For successful establishment of desired plants on a recent disturbance, knowing species composition in the soil propagule bank may help predict which species survive and which may become a management problem. The amount and composition of viable soil propagules in donor soils will initially determine the trajectory of early succession on reclaimed lands.

Implications for Practice

- Salvaged soils can harbor appropriate plant species to develop diverse, self-sustaining ecosystems that existed prior to disturbance for reclaimed landscapes.
- LFH provides more appropriate plant propagules than peat for boreal upland forest sites.
- Selection of a donor soil should include consideration of added soil nutrients, propagule availability, and woody debris in addition to organic matter.
- Knowledge of the composition and abundance of species in soil propagule banks can be applied to determine if species need to be added or controlled, regardless of the location or type of disturbance.

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